

DBT Exam

Name: Aditya Ingale
Roll no.: 260250320005

Q1.

Ans:

```
CREATE TABLE Category (  
    category_id INT,  
    category_name VARCHAR(50)  
);
```

```
CREATE TABLE Medicines (  
    medicine_id INT,  
    medicine_name VARCHAR(100),  
    price DECIMAL(10,2),  
    stock_qty INT,  
    category_id INT  
);
```

```
CREATE TABLE Sales (  
    sale_id INT,  
    medicine_id INT,  
    quantity INT,  
    sale_date DATE  
);
```

```
ALTER TABLE Category  
    ADD PRIMARY KEY (category_id);
```

```
ALTER TABLE Medicines  
    ADD PRIMARY KEY (medicine_id);
```

```
ALTER TABLE Sales  
    ADD PRIMARY KEY (sale_id);
```

```
ALTER TABLE Medicines  
    ADD CONSTRAINT fk_medicines_category  
    FOREIGN KEY (category_id) REFERENCES Category(category_id);
```

```
ALTER TABLE Sales  
    ADD CONSTRAINT fk_sales_medicine  
    FOREIGN KEY (medicine_id) REFERENCES Medicines(medicine_id);
```

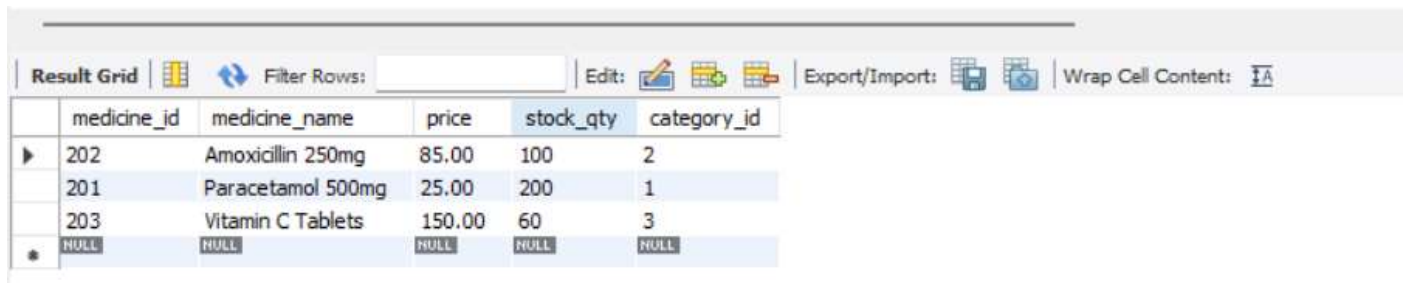
```
INSERT INTO Category (category_id, category_name) VALUES  
(1, 'Pain Relief'),  
(2, 'Antibiotic'),  
(3, 'Vitamins');
```

```
INSERT INTO Medicines (medicine_id, medicine_name, price, stock_qty, category_id) VALUES  
(201, 'Paracetamol 500mg', 25.00, 200, 1),  
(202, 'Amoxicillin 250mg', 85.00, 100, 2),  
(203, 'Vitamin C Tablets', 150.00, 60, 3);
```

```
INSERT INTO Sales (sale_id, medicine_id, quantity, sale_date) VALUES
```

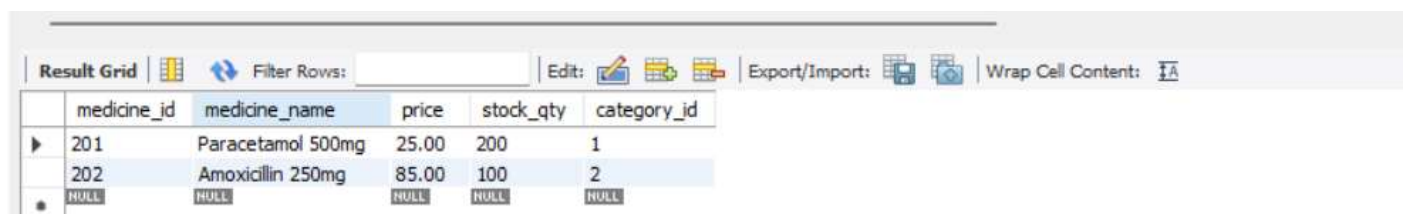
```
(1, 201, 3, '2026-07-01'),  
(2, 202, 1, '2026-07-02'),  
(3, 203, 2, '2026-07-03');
```

```
SELECT * FROM Medicines  
ORDER BY medicine_name ASC;
```



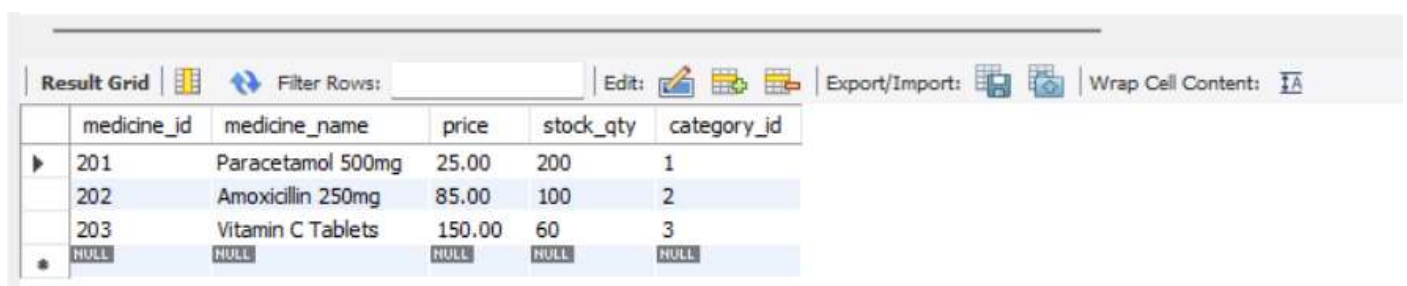
	medicine_id	medicine_name	price	stock_qty	category_id
▶	202	Amoxicillin 250mg	85.00	100	2
	201	Paracetamol 500mg	25.00	200	1
	203	Vitamin C Tablets	150.00	60	3
*	NULL	NULL	NULL	NULL	NULL

```
SELECT * FROM Medicines  
WHERE category_id = 1 OR category_id = 2;
```



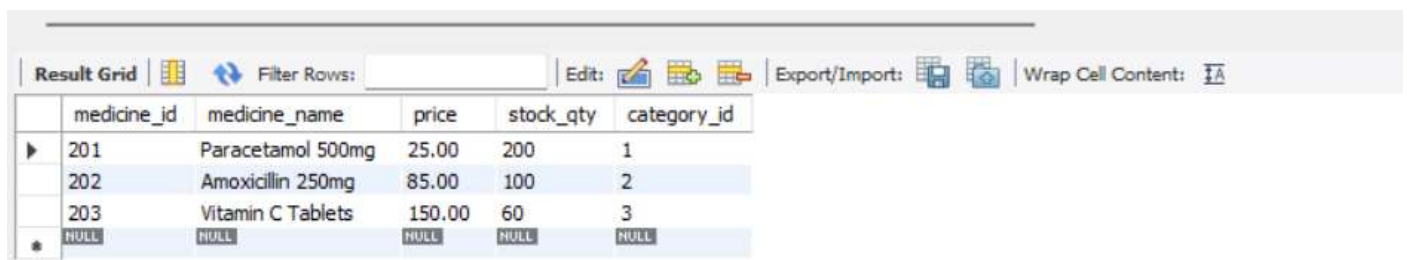
	medicine_id	medicine_name	price	stock_qty	category_id
▶	201	Paracetamol 500mg	25.00	200	1
	202	Amoxicillin 250mg	85.00	100	2
*	NULL	NULL	NULL	NULL	NULL

```
SELECT * FROM Medicines  
LIMIT 5;
```



	medicine_id	medicine_name	price	stock_qty	category_id
▶	201	Paracetamol 500mg	25.00	200	1
	202	Amoxicillin 250mg	85.00	100	2
	203	Vitamin C Tablets	150.00	60	3
*	NULL	NULL	NULL	NULL	NULL

```
SELECT * FROM Medicines  
WHERE price BETWEEN 20 AND 150;
```



	medicine_id	medicine_name	price	stock_qty	category_id
▶	201	Paracetamol 500mg	25.00	200	1
	202	Amoxicillin 250mg	85.00	100	2
	203	Vitamin C Tablets	150.00	60	3
*	NULL	NULL	NULL	NULL	NULL

```
SELECT category_id, COUNT(*) AS total_medicines, SUM(stock_qty) AS total_stock  
FROM Medicines  
GROUP BY category_id;
```

Result Grid			
	category_id	total_medicines	total_stock
▶	1	1	200
	2	1	100
	3	1	60

```
SELECT Medicines.medicine_name, Sales.quantity, Sales.sale_date
FROM Medicines
LEFT JOIN Sales ON Medicines.medicine_id = Sales.medicine_id;
```

Result Grid			
	medicine_name	quantity	sale_date
▶	Paracetamol 500mg	3	2026-07-01
	Amoxicillin 250mg	1	2026-07-02
	Vitamin C Tablets	2	2026-07-03

```
CREATE TEMPORARY TABLE Temp_Medicines AS
SELECT * FROM Medicines WHERE stock_qty > 50;
```

```
SELECT * FROM Temp_Medicines;
```

```
DROP TEMPORARY TABLE Temp_Medicines;
```

Result Grid					
	medicine_id	medicine_name	price	stock_qty	category_id
▶	201	Paracetamol 500mg	25.00	200	1
	202	Amoxicillin 250mg	85.00	100	2
	203	Vitamin C Tablets	150.00	60	3

```
DELIMITER //
```

```
CREATE PROCEDURE GetStockByMedicine(
    IN p_medicine_id INT,
    OUT p_stock INT
)
BEGIN
    SELECT stock_qty INTO p_stock
    FROM Medicines
    WHERE medicine_id = p_medicine_id;
END //
```

```
DELIMITER ;
```

```
-- To call the procedure:
CALL GetStockByMedicine(201, @stock);
SELECT @stock;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
@stock			
200			

DELIMITER //

```
CREATE TRIGGER before_medicine_update
BEFORE UPDATE ON Medicines
FOR EACH ROW
BEGIN
    IF NEW.stock_qty < 0 THEN
        SET NEW.stock_qty = 0;
    END IF;
END //
```

DELIMITER ;

Q2.

Ans:

use medicalstore

```
db.createCollection("medicines")
```

```
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a70f18fe3d2c6f717bb8964'),
    '1': ObjectId('6a70f18fe3d2c6f717bb8965'),
    '2': ObjectId('6a70f18fe3d2c6f717bb8966')
  }
}
```

```
db.medicines.insertMany([
  { medicine_id: 201, medicine_name: "Paracetamol 500mg", price: 25, category: "Pain Relief", stock_qty: 200 },
  { medicine_id: 202, medicine_name: "Amoxicillin 250mg", price: 85, category: "Antibiotic", stock_qty: 100 },
  { medicine_id: 203, medicine_name: "Vitamin C Tablets", price: 150, category: "Vitamins", stock_qty: 60 }
])
```

```

< {
  _id: ObjectId('6a70f18fe3d2c6f717bb8964'),
  medicine_id: 201,
  medicine_name: 'Paracetamol 500mg',
  price: 25,
  category: 'Pain Relief',
  stock_qty: 200
}
{
  _id: ObjectId('6a70f18fe3d2c6f717bb8965'),
  medicine_id: 202,
  medicine_name: 'Amoxicillin 250mg',
  price: 85,
  category: 'Antibiotic',
  stock_qty: 100
}
{
  _id: ObjectId('6a70f18fe3d2c6f717bb8966'),
  medicine_id: 203,
  medicine_name: 'Vitamin C Tablets',
  price: 150,
  category: 'Vitamins',
  stock_qty: 60
}

```

db.medicines.find()

```

db.medicines.find({
  $or: [
    { price: { $gt: 200 } },
    { category: "Pain Relief" }
  ]
})

```

```

< {
  _id: ObjectId('6a70f18fe3d2c6f717bb8964'),
  medicine_id: 201,
  medicine_name: 'Paracetamol 500mg',
  price: 25,
  category: 'Pain Relief',
  stock_qty: 200
}
>

```

```
db.medicines.find().sort({ price: -1 })
```

```
< {
  _id: ObjectId('6a70f18fe3d2c6f717bb8966'),
  medicine_id: 203,
  medicine_name: 'Vitamin C Tablets',
  price: 150,
  category: 'Vitamins',
  stock_qty: 60
}
{
  _id: ObjectId('6a70f18fe3d2c6f717bb8965'),
  medicine_id: 202,
  medicine_name: 'Amoxicillin 250mg',
  price: 85,
  category: 'Antibiotic',
  stock_qty: 100
}
{
  _id: ObjectId('6a70f18fe3d2c6f717bb8964'),
  medicine_id: 201,
  medicine_name: 'Paracetamol 500mg',
  price: 25,
  category: 'Pain Relief',
  stock_qty: 200
}
```

```
db.medicines.createIndex({ medicine_name: 1 })
```